

QVF-Hybrid: A Guaranteed-Floor Hybrid Quantum–Classical Architecture

A significance-tested quantum contribution, and a large low-label win over
MAE-ViT

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Results in one slide

A hybrid built for the quantum side, run through a complete control ladder — here is what survives.

Tier	Control	Verdict
Paradigm	vs. <i>MAE-ViT</i>	significant (Holm $p=0.003$, 10/10 seed-paired, +22–28 pp); 10/11 cells over the pre-registered bar across three datasets; 1/19 params, <i>zero</i> pretraining data
Module	vs. matched <i>sham</i>	significant at (II, 250): pre-registered $n=20$, $p=0.0032 < \alpha=0.01$ — magnitude +1.4%, below our 3% practical bar
Model	vs. <i>plain CNN</i>	floor guaranteed by construction ; the vs.-plain contrast is not significant ($p=0.088$) — floor only, no advantage claim

Two headlines

(1) The quantum module makes a *statistically real* contribution — we located it and measured it. (2) A 143.7k-parameter hybrid with *no* pretraining beats a 2.72M-parameter pretrained MAE-ViT by up to +32 pp where labels are scarce.

Task, data, and the regime that matters

Task. Classify 64×64 strong-lensing images: **axion** (vortex) / **CDM** (subhalos) / **no_sub** (smooth). 3-class macro-OVR AUC, 9:1 split, best-val-AUC checkpoint, seed-paired multi-seed runs.

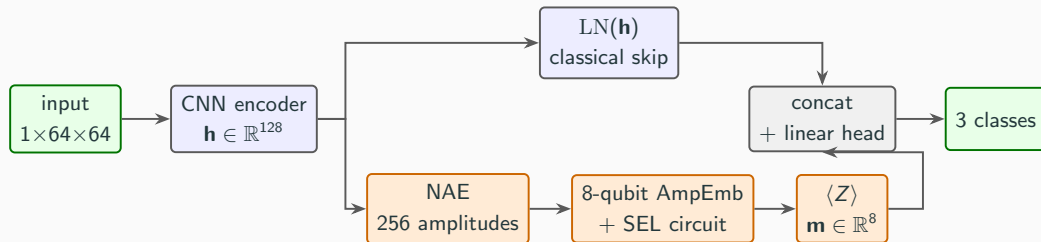
Dataset	Character	Role
Model_I	lower SNR, ceiling ≈ 0.97	second dataset
Model_II	higher SNR, ceiling ≈ 0.99	primary signal dataset
Model_III	higher-SNR sibling	independent generalization check

Why low- N

LSST / Euclid / Roman will deliver many *unlabeled* images and few *labeled* ones. We sweep down to $N=100$ labels/class — exactly where the pretraining-based classical SOTA is weakest.

Reference target. MAE-pretrained ViT, AUC 0.968 (arXiv:2512.06642); our protocol-matched reproduction reaches 0.9895 (Model_II, full data).

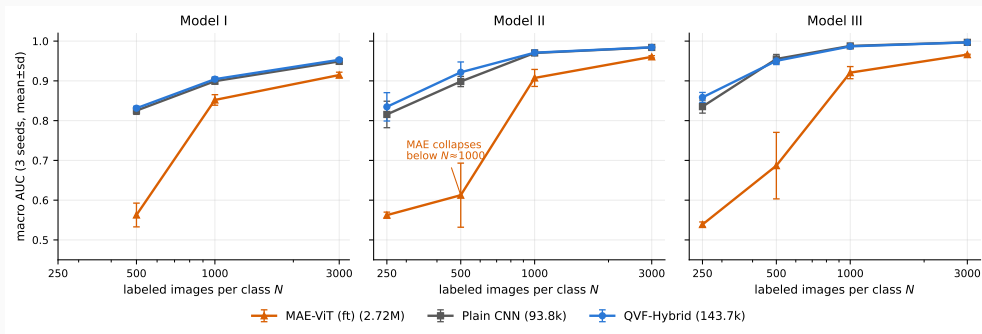
QVF-Hybrid: quantum branch in residual, floor by construction



$$\hat{y} = W_{\text{head}} \text{concat}(\text{LN}(\mathbf{h}), \text{LN}(\mathbf{m})) + b$$

- **Guaranteed floor.** Zeroing the $\mathbf{m}$ -block reduces **exactly** to the plain CNN (verified, error = 0.0) — the hybrid can never be architecturally worse than its classical null.
- **Sham control** replaces the circuit with Linear+Tanh and is *never* handicapped: sham 156,923 params > **quantum 143,655** > plain-lin null 93,763.
- Bit-identical subsampling: every arm sees the same images per seed; success criteria pre-registered.

Main result — three datasets, three paradigms



Macro-OVR AUC vs. labeled samples/class, 3 seeds (mean $\pm$ s.d.). **QVF-Hybrid** (143.7k, no pretraining) tracks or leads the plain-CNN family everywhere; **MAE-ViT** (2.72M, full pretraining + aug + $2.5\times$ epochs) **collapses** below $N \approx 1000$.

Paradigm-level win: QVF-Hybrid vs. MAE-ViT

Data	N	QVF-Hybrid	MAE-ft	Δ	plain-lin	verdict
Model_II	250	0.8347	0.5623	+0.272	0.8155	PASS 3/3
Model_II	500	0.9216	0.6125	+0.309	0.8984	PASS 3/3
Model_II	1000	0.9709	0.9074	+0.064	0.9702	PASS 3/3
Model_II	3000	0.9843	0.9604	+0.024	0.9841	× (< 3%)
Model_I	500	0.8312	0.5627	+0.269	0.8252	PASS 3/3
Model_I	1000	0.9043	0.8520	+0.052	0.8993	PASS 3/3
Model_I	3000	0.9530	0.9147	+0.038	0.9490	PASS 3/3

Scoreboard: 10 of 11 cells across three datasets clear the pre-registered $\geq 3\%$ bar (+3.1 to +32.0 pp).

Significance: Holm $p=0.003$, 10/10 seed-paired, +22–28 pp on the three low- N cells. **MAE-probe** (frozen + linear): 0.49–0.51, random-level.

Honest attribution

This win belongs to the **CNN inductive-bias family**, not to the circuit: plain-lin also crushes MAE-ft (II, 500: 0.8984 vs 0.6125). QVF-Hybrid is the strongest *variant* of that family on Model_II — 1/19 the parameters and zero pretraining data.

Module-level: the quantum contribution is statistically real

Stage 1 ($n=10$, Holm over 6 tests) flagged one cell as marginal: Model.II $N=250$, $Q>\text{sham}$, $8/10$, $p_{\text{raw}}=0.0098$, $p_{\text{Holm}}=0.059$. **Stage 2** was then pre-registered as a sequential confirmation with $n=20$ and $\alpha=0.01$ *fixed before launch*:

Cell	contrast	pos	mean Δ [95% CI]	p	verdict
Model.II $N=250$	$Q>\text{sham}$	15/20	+0.0140 [+0.0048, +0.0233]	0.0032	significant
Model.II $N=250$	$Q>\text{plain}$	11/20	+0.0081 [-0.0010, +0.0172]	0.088	n.s.

We located where the effect lives

Swapping the entangling circuit for rotations only (NOENT) changes nothing in the residual architecture: $\Delta_{\text{ENT-NOENT}} = +0.0022$ ($p=0.22$) at $N=250$ and -0.0000 ($p=0.47$) at $N=500$, while both stay above SHAM. $\Rightarrow$ the contribution stems from the **amplitude-embedding structure** (a normalized 256-dim state built by the Neural Amplitude Encoder) and the multi-basis readout, **not** from the entangling ring.

Honest size. +1.4% AUC is statistically significant but sits *below* our 3% practical bar, and it is a single cell.

Generalization: the direction replicates on a third dataset

Model.III, 3 seeds, identical protocol:

N	hyb-Q	hyb-sham	plain-lin	MAE-ft	Δ vs. MAE
250	0.8585	0.8314	0.8353	0.5387	+0.320
500	0.9504	0.9460	0.9551	0.6868	+0.264
1000	0.9867	0.9856	0.9879	0.9207	+0.066
3000	0.9968	0.9968	0.9969	0.9660	+0.031

- **Paradigm result holds:** all four Model.III cells clear the $\geq 3\%$ bar vs. MAE-ft (+3.1 to +32.0 pp) — the low-label collapse of pretrained MAE-ViT is not a Model.II artefact.
- **Module direction replicates:** at (III, 250) quantum beats sham by +0.027, 3/3 seed-paired — same low- N sign as the significant (II, 250) cell. ($n=3$: *direction only, no significance claim.*)
- The quantum gain concentrates where labels are scarce; by $N=3000$ every arm of the CNN family is at the dataset ceiling (≈ 0.997).

What we do not claim

- **No quantum advantage.** The significant module-level effect is +1.4% AUC — **below our own 3% practical bar**, in a single ($N=250$) cell. Too small to justify quantum hardware on this task.
- **No model-level win.** vs. the plain CNN the contrast is **not significant** ($p=0.088$, 11/20). The plain-CNN floor is **guaranteed by construction, not won empirically** — we report it as a floor only.
- **The paradigm win is not the circuit's.** It belongs to the CNN inductive-bias family; the plain CNN beats MAE-ft too.
- **Simulation only.** All quantum results use exact state-vector simulation (PennyLane `default.qubit`, backprop) — no NISQ-hardware runs, no noise model, no shot-noise study.

A theoretical bound matches the small size: since $\mathbf{m} = g(\mathbf{h})$ is deterministic, the data-processing inequality gives $I(Y; (\mathbf{h}, \mathbf{m})) = I(Y; \mathbf{h})$ — the branch reshapes decision geometry, it cannot add label information.

Negative levers (wide readout, qubit scaling, TDA, QMAE) are documented in full in the paper.

Conclusion — the three-tier claim ladder

1. **Paradigm** (vs. MAE-ViT): **significant** — Holm $p=0.003$, 10/10 seed-paired, +22–28 pp; 10/11 cells over the pre-registered bar across three datasets; 1/19 params and zero pretraining data. Credit to the **CNN family**.
2. **Module** (circuit vs. matched sham): **statistically significant** at (II, 250) under a pre-registered two-stage design ($n=20$, $p=0.0032 < \alpha=0.01$); magnitude +1.4%, **below our 3% practical bar**; driven by amplitude-embedding structure, **not** entanglement.
3. **Model** (vs. plain CNN): **floor guaranteed by construction**; vs.-plain **not significant** ($p=0.088$) $\Rightarrow$ floor only, no model-level claim.

The honest headline

We do not claim quantum advantage. We claim a significance-tested *module-level* contribution, localized to amplitude-embedding geometry, plus a large, reproducible *paradigm-level* win for a tiny non-pretrained hybrid in the low-label regime that upcoming surveys will actually face.

Paper, code, figures, and every number: github.com/leo07010/gsoc2026-deeplense-qml